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People Identification via Face Recognition

AI Computer Vision Project

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1. Introduction

This project addresses the problem of people identification in photographs using a face recognition pipeline built on top of DeepFace, a deep learning facial recognition system. The system is designed to identify individuals appearing in scene images by comparing detected faces against a pre-built gallery of known identities.

The core challenge is not merely detecting faces, but accurately matching them to a gallery under realistic conditions - including variable lighting, pose and notably, image blurriness. The project implements a multi-gate gallery matcher, evaluates it on a dataset of scene images across all identities, and performs a systematic grid search across detector-recognizer combinations to find the optimal configuration.

2. Dataset and Setup

2.1 Dataset Structure

The dataset used was from DroneFace (https://hjhsu.github.io/DroneFace/), an open dataset for testing face recognition from drone and scene imagery. It is organized under open_data_set/. The two experimental notebooks mainly use photos_all_faces/ and photos_all/; portraits/ is included only as supporting/illustrative data and is not used in the reported voting or grid-search metrics:

- photos_all_faces/ - consists of 1,364 pre-cropped face images, with 124 images per identity for each of the 11 identities (A–K). These are tight face crops used to build the recognition gallery.
- photos_all/ - a set of 620 real-world photos taken in outdoor and indoor environments. Each scene filename encodes the ground-truth identities present via a token prefix (e.g., cd_gp_0_eo_12.JPG → identities C and D are expected in the scene).
- portraits/ - contains 7 portrait images with subject IDs A, D, E, G, I, J, and K. This folder is useful for visual reference, but it is not used to build the gallery or compute the final evaluation results.

Fig. 1 tight face crops from photos_all_faces/ (Id: A)

Fig. 2 photo from photos_all/

Fig. 3: Portrait photo from portraits/ (ID: A)

2.2 Dependencies and Environment

The project runs in a Python 3.9.6 virtual environment (.venv) with the following key libraries: DeepFace (face detection, alignment, and embedding), TensorFlow/tf-keras (backend for deep learning models), OpenCV (image I/O and manipulation), pandas and NumPy (data handling), matplotlib (visualization) and retina-face for the RetinaFace detector experiment.

3. System Architecture

3.1 Pipeline Overview

The identification pipeline follows a standard gallery-based face recognition flow:

```
Face Detection  →  Face Cropping & Alignment  →  Embedding Extraction  →
Gallery Voting  →  Identity Decision
```

3.2 Gallery Construction

In the main full-evaluation notebook, a gallery DataFrame is built by iterating through all 1,364 face images in photos_all_faces/. Using Facenet512, the selected recognizer from the grid search, each gallery crop is passed through DeepFace.represent() to extract a 512-dimensional face embedding. The resulting embeddings are stored as float32 NumPy arrays alongside the identity label inferred from the filename. In the detector-recognizer grid-search notebook, this gallery-building step is repeated once per recognizer model: GhostFaceNet, Facenet512, and ArcFace.

3.3 Scene Processing

For each scene image, the pipeline first runs DeepFace.extract_faces() using MTCNN as the fixed extractor. Each detected face is validated (minimum 10×10 pixels) and cropped. The crop is then passed to DeepFace.represent() to extract an embedding. That embedding is then matched against the gallery using the voting mechanism explained in Section 3.4.

3.4 Multi-Gate Gallery Matcher

The core matching algorithm is a multi-gate voting system based on cosine distance. Each detected face crop is embedded, compared against the gallery, and accepted only if the winning identity passes the vote and distance gates. The cosine distance is computed as:

```
distance(a, b) = 1 − (a · b) / (‖a‖ · ‖b‖)
```

For a given probe embedding, the code first counts gallery rows with cosine distance less than or equal to MATCH_THRESHOLD. In the full Facenet512 evaluation, MATCH_THRESHOLD = 0.55. Each row under the threshold casts one vote for its identity. The program keeps only identities with at least MIN_VOTES votes, then applies two extra gates: mean voting distance must be <= MEAN_VOTE_DIST_MAX and best/minimum voting distance must be <= MIN_BEST_VOTE_DIST_MAX. Among the remaining candidates, the winner is the identity with the lowest mean voting distance, using lower minimum distance and higher vote count as tie-breakers. This is implemented in the full evaluation notebook (people_identification_voting_simple.ipynb).

3.5 Evaluation Metrics

Performance is measured using a set-based formulation per scene:

- TP = |expected ∩ predicted| — correctly identified token identities
- FP = |predicted − expected| — identities predicted but not in the token
- FN = |expected − predicted| — token identities not recovered by the system
- Precision = TP / (TP + FP); Recall = TP / |expected|
- Score = 0.5 × mean_precision + 0.5 × mean_recall (used for grid search)
- token_ok = True if FP = 0 and FN = 0 (strict exact match)

Rejected faces (no accepted match) do not contribute to FP. Only accepted predictions that fall outside the expected set are penalized as false positives.

4. Experiments

4.1 Simple Voting Evaluation (Full Dataset)

The baseline configuration uses MTCNN as the detector, Facenet512 as the recognition model, with MATCH_THRESHOLD = 0.55, MIN_VOTES = 20, MEAN_VOTE_DIST_MAX = 0.48, and MIN_BEST_VOTE_DIST_MAX = 0.37.This configuration was evaluated across all 620 scenes in photos_all/.

Full Evaluation Results (620 Scenes)

Metric	Value
Scenes Evaluated	620
Scenes with Strict Token Match (token_ok)	248 / 620 (40.0%)
Mean Precision jt	0.7785
Mean Recall	0.5901
Mean F1 Score	~0.671 (estimated)

The system exactly recovered the expected identity set with no false positives in 248 out of 620 scenes. The notebook mean precision (0.7785) is higher than mean recall (0.5901), indicating that the system is conservative: when it accepts a match it is usually correct, but it rejects or misses many true identities. Precision is undefined for scenes with no accepted predictions, so the reported mean precision skips NaN values; the F1 value above treats NaNs as 0.

4.2 Demo Scene Analysis

A qualitative demo was run on the scene cd_gp_0_eo_12.JPG (expected identities: C and D). MTCNN detected 4 faces in the scene. The results were:

Face	Predicted	Accepted	Votes	Mean Vote Dist.	Min Vote Dist.
Face 0	---	No	56	0.4934	0.4067
Face 1	J	Yes	87	0.3375	0.2312
Face 2	D	Yes	84	0.4707	0.3381
Face 3	C	Yes	23	0.4788	0.3369

Face 0 was rejected because it won 56 votes for identity A, but A's mean voting distance (0.4934) exceeded the MEAN_VOTE_DIST_MAX gate of 0.48. Faces 2 and 3 correctly matched D and C. Face 1 was predicted as J — a false positive, as J is not in the expected token {C, D}. This reflects one of the system's main failure modes: occasionally accepting a stray person visible in the background or a person whose embedding falls within the threshold of a gallery identity.

4.3 Detector x Recognizer Grid Search (15 Scenes)

A systematic grid search was conducted across 2 detectors (MTCNN, RetinaFace) and 3 recognition models (GhostFaceNet, Facenet512, ArcFace), evaluating 10 cosine-distance thresholds per combination (T in {0.08, 0.10, 0.12, 0.15, 0.18, 0.22, 0.28, 0.35, 0.45, 0.55}) on the first 15 scenes. The grid search used a simplified matcher with MIN_VOTES = 5 and no mean/min distance gates; after candidates pass the vote floor, the winner is still selected by lowest mean voting distance.

Best Result per Detector × Model (after threshold tuning)

Detector	Model	Best Threshold	Score	Mean F1	Token OK Rate
mtcnn	Facenet512	0.55	0.317	0.300	13.3%
mtcnn	ArcFace	0.08	0.050	0.044	0.0%
mtcnn	GhostFaceNet	0.08	0.000	0.000	0.0%
retinaface	ArcFace	0.08	0.000	0.000	0.0%
retinaface	Facenet512	0.08	0.000	0.000	0.0%
retinaface	GhostFaceNet	0.08	0.000	0.000	0.0%

The overall winner of the grid search is MTCNN + Facenet512 at threshold T = 0.55, achieving a score of 0.317 and a token_ok rate of 13.3% on the 15-scene subset. All RetinaFace combinations scored zero in this small experiment. This does not prove that RetinaFace is generally worse; it means that under this notebook's settings and first-15-scene subset, RetinaFace produced no correct accepted token matches. The runtime difference is still useful context: MTCNN required about 12-24 minutes per recognizer sweep, while RetinaFace completed in about 26-28 seconds per recognizer sweep, which is consistent with fewer usable detections or accepted crops in this setup.

5. Results Discussion

5.1 Impact of the Filtering Mechanism

The mean and minimum distance gates (MEAN_VOTE_DIST_MAX and MIN_BEST_VOTE_DIST_MAX) make the full evaluator more conservative than a vote-floor-only matcher. Without these gates, a face can receive many votes for an identity even when those votes are spread across relatively high distances, meaning the match is uncertain. The demo scene shows this clearly: Face 0 receives 56 votes for identity A, but it is rejected because its mean voting distance is 0.4934, above the 0.48 gate. These gates reduce spurious identifications, but they also reduce recall because some true faces are rejected.

5.2 Why Performance Does Not Exceed ~78% Precision / ~59% Recall

Despite the filtering improvements we tried to add to improve performance, the system does not reach the expected 90%+ accuracy. Several interrelated factors explain this:

Pre-trained Models Without Fine-Tuning

DeepFace provides access to pre-trained recognition models such as Facenet512, ArcFace, and GhostFaceNet. These models were not fine-tuned or threshold-calibrated on this project's specific DroneFace subset. Fine-tuning or calibration could help separate the 11 project identities (A-K), but it was not performed in these notebooks. With only 124 gallery crops per identity and no separate training/validation split used for model training, the system relies on general-purpose embeddings and manually selected thresholds rather than identity-specialized recognition.

Image Quality — Blur and Low Resolution

Scene images exhibit significant blurriness, which is a major source of degraded recognition performance. Face recognition CNNs rely on fine-grained spatial features — the precise texture of skin, the sharpness of facial contours, the detail of eye and nose shape — to discriminate between individuals. When faces are blurry (due to motion, camera focus, or distance from camera), these discriminative features are lost or averaged out. The resulting face crops that enter the recognizer contain less information, pushing embeddings of different people closer together in the latent space and making reliable separation harder.

Demographic and Distribution Mismatch

Another possible factor is distribution mismatch. The recognition models were trained on large public face datasets that may differ from DroneFace in camera angle, subject distance, resolution, blur, lighting, demographics, and drone/scene conditions. Because the project did not fine-tune or calibrate the models on this exact domain, some identities may map close together in embedding space, making threshold-based separation difficult. This should be treated as a plausible explanation rather than a measured conclusion from the two notebooks.

6. Key Findings and Conclusions

The project successfully implements a complete people identification pipeline using voting-based gallery matching with DeepFace. The main findings are:

- Within the 15-scene grid search, MTCNN + Facenet512 at threshold T = 0.55 was the best setting tested, achieving a mean score of 0.317 and a token_ok rate of 13.3%. The same detector-recognizer setting was then used in the full simple-voting evaluation, producing 0.7785 mean precision and 0.5901 mean recall over 620 scenes.
- RetinaFace underperformed in this specific 15-scene grid-search setup, achieving zero score across the tested recognizer combinations. The result should be interpreted as setup- and subset-specific, not as a general conclusion about RetinaFace.
- The filtering mechanism (mean and minimum distance gates) helped make the final evaluator conservative by rejecting geometrically ambiguous matches, such as detections with many votes but high average voting distance.
- Pre-trained models without domain-specific fine-tuning or calibration, combined with blurry/low-resolution scene crops and possible dataset distribution mismatch, are the main bottlenecks for performance.
- 248 out of 620 scenes (40%) achieved a strict token match: zero false positives and zero false negatives. This shows that the system performs well in favorable conditions but struggles with blur, missed detections, extra background faces, and hard identity separations.

Future improvements could include creating a validation split for threshold calibration, fine-tuning or training a lightweight classifier on top of the embeddings, using deblurring or super-resolution preprocessing for low-quality crops, expanding the grid search beyond the first 15 scenes, and testing ensemble voting across multiple recognizers.

7. Contributions

Hassan Eltahir - Dataset management and gallery embedding
Maher Abdul Gafoor - Scene demo, Full Evaluation
Mohammed Shamiul Islam - Multi-gate gallery matcher logic
Zayed Muhammad Shakir Khan - Experimenting with different detectors and recognizers

8. Appendix — Hyperparameter Summary

Parameter	Value	Description
MODEL_NAME	Facenet512	Recognition model (best in grid search)
DETECTOR_BACKEND	mtcnn	Face detector (best in grid search)
MATCH_THRESHOLD (T)	0.55	Max cosine distance for a gallery row to cast a vote
MIN_VOTES	20	Minimum votes for an identity to be a candidate (simple eval)
MIN_VOTES (grid search)	5	Reduced vote floor for the grid search experiment
MEAN_VOTE_DIST_MAX	0.48	Max mean voting distance for a candidate to be accepted
MIN_BEST_VOTE_DIST_MAX	0.37	Max minimum voting distance for a candidate to be accepted
Gallery Size	1,364 images	11 identities × 124 images each
Scene Set Size	620 scenes	Full evaluation set
Threshold Grid	0.08 – 0.55 (10 values)	Cosine distance values swept in grid search